Newton School of Technology

Data Visualization & Analytics

Retail Analytics: End-to-End Data Pipeline & Interactive Dashboard

Final Project Report

Field	Details
Sector	Retail & E-Commerce
Team ID	SectionB Group 3
Team Members	Bulbul Agarwalla, Shourya Pratap, Varun Udata, Nishant Sharma, Anant Lalwani, Parul
Roles	Data Engineer, Analyst, Visualization Specialist
Faculty Mentor	Please add mentor name
Institute	Newton School of Technology
GitHub Repository	https://github.com/Twinklingstar374/SectionB_G3_RetailAnalytics.git
Tableau Dashboard	Please add Tableau Public URL
Submission Date	April 29, 2026

1. Executive Summary

Problem

The retail sector increasingly depends on data-driven decision-making to remain competitive. This project addresses the challenge of transforming 302,010 raw retail transactions into actionable intelligence for business leaders. Without proper analysis, key patterns in customer behaviour, product performance, and operational efficiency remain hidden, leading to suboptimal business decisions.

Approach

We built an end-to-end analytics pipeline using Python and Tableau. Raw data was extracted, cleaned, and engineered across five Jupyter notebooks, producing a tableau-ready dataset of 30+ columns. Four interactive Tableau dashboards were then built to support executive and operational decision-making.

Key Insights

- Regular customers drive 48% of total revenue, making retention of this segment the highest-priority business action.
- April is the peak revenue month generating Rs. 57 Cr, while November/December records the lowest revenue, indicating seasonal patterns requiring targeted promotions.
- Same-Day shipping generates the highest revenue (Rs. 142Cr) and is the most preferred method, suggesting investment in fast delivery infrastructure.
- 94.7% order success rate indicates strong operational performance, with the remaining 5.3% representing recoverable revenue through process improvement.

Key Recommendations

- Launch a Regular Customer Loyalty Programme to increase retention and grow segment revenue beyond Rs. 200Cr.
- Introduce November/December promotional campaigns to address seasonal revenue dip and recover estimated Rs. 8-10Cr in lost revenue.
- Expand Same-Day delivery capacity to all regions to capitalise on the highest-revenue shipping preference.

2. Sector & Business Context

The global retail and e-commerce sector is undergoing rapid transformation driven by digital adoption, changing consumer preferences, and increasing competition from online marketplaces. Businesses that leverage data analytics gain a significant competitive advantage in understanding customer behaviour, optimising inventory, and improving operational efficiency.

The decision-makers this project serves include Chief Revenue Officers who need visibility into sales trends, Chief Marketing Officers who require customer segmentation intelligence, and Operations Managers who need real-time order fulfilment metrics. The project was chosen because retail transaction data contains rich, multi-dimensional insights that remain unexploited without a structured analytics pipeline.

The business value of solving this problem is substantial: improved customer targeting can increase conversion rates by 15-20%, operational efficiency gains from order status monitoring can reduce cancellations by 5-8%, and data-driven product decisions can optimize inventory costs by 10-15%.

3. Problem Statement & Objectives

Formal Problem Definition

How can a retail organisation transform 302,010 raw transactional records into a structured, interactive analytics system that enables data-driven decision-making across sales, customer, product, and operational dimensions?

Project Scope

- In scope: Data extraction, cleaning, EDA, statistical analysis, KPI computation, and Tableau dashboard development.
- In scope: Customer segmentation, product performance analysis, revenue trend analysis, and operational insights.
- Out of scope: Real-time data ingestion, predictive machine learning models, and CRM system integration.

Success Criteria

- A clean, analysis-ready dataset with zero null values committed to GitHub.

- Five documented Jupyter notebooks covering the full analytics pipeline.
- Four interactive Tableau dashboards published on Tableau Public with working filters.
- Actionable insights and recommendations supported by statistical evidence.

4. Data Description

Dataset Overview

Property	Value
Dataset Name	new_retail_data_2.csv
Total Rows	302,010 transactions
Total Columns	30+ (25 original + 9 engineered)
Time Period	Full calendar year (Jan-Dec)
Geography	USA, UK, Germany, Australia, Canada
File Size	~45MB

Key Columns

Column	Type	Description
Transaction_ID	String	Unique transaction identifier
Customer_ID	String	Unique customer identifier
Age	Integer	Customer age in years
Gender	String	Male / Female
Income	String	Low / Medium / High / Premium
Customer_Segment	String	New / Regular / Premium
Total_Amount	Float	Total transaction value in Rs.
Product_Category	String	Product category name
Order_Status	String	Delivered / Shipped / Processing / Pending
Ratings	Float	Customer rating 1 to 5
Shipping_Method	String	Same-Day / Express / Standard
Payment_Method	String	Cash / Credit Card / Debit Card / PayPal

Data Limitations

- No timestamp granularity beyond day level, limiting intra-day analysis.
- Geographic data limited to country level; city-level analysis may be incomplete.

- PII columns (Name, Email, Phone, Address, Zipcode) were removed for privacy compliance, limiting customer-level personalisation analysis.

5. Data Cleaning & ETL Pipeline

All cleaning and transformation steps were executed in Python across notebooks/02_cleaning.ipynb, with the pipeline encapsulated in scripts/etl_pipeline.py.

Missing Value Treatment

Column	Nulls	Treatment Method
Age	~180	Filled with median age
Amount	~180	Filled with median amount
Ratings	184	Filled with median rating
Gender	~180	Filled with mode (most frequent)
Customer_Segment	~180	Filled with mode
City / State / Country	~180	Filled with 'Unknown'
Date	~180	Forward filled (ffill)
Transaction_ID / Customer_ID	~180	Forward filled (ffill)

Feature Engineering

Nine new columns were engineered to support dashboard KPIs and statistical analysis:

Column	Logic	Business Purpose
avg_order_value	Total_Amount / Total_Purchases	Measures spend per item
age_group	pd.cut(Age, bins=[0,25,35,45,60,100])	Enables age-based segmentation
purchase_category	pd.cut(Total_Purchases, bins)	Classifies purchase frequency
revenue_category	pd.cut(Total_Amount, bins)	Revenue tier classification
weekday	Date.dt.day_name()	Day of week analysis
is_weekend	Date.dt.dayofweek >= 5	Weekend vs weekday comparison
order_success	Order_Status == 'Delivered'	Binary success flag for KPI
rating_category	Based on Ratings score	Qualitative rating label
customer_value_segment	Based on Total_Amount	Value-based segmentation

6. KPI & Metric Framework

KPI	Formula	Value	Business Relevance
Total Revenue	SUM(Total_Amount)	Rs. 413Cr	Primary business health indicator
Total Orders	COUNT(Transaction_ID)	2,94,461	Volume of business activity
Avg Order Value	AVG(avg_order_value)	Rs. 1,404	Customer spending efficiency
Avg Rating	AVG(Ratings)	3.163 / 5	Customer satisfaction benchmark
Total Unique Customers	COUNTD(Customer_ID)	86,766	Customer base size
Order Success Rate	SUM(order_success)/COUNT(*)	94.7%	Operational efficiency
Total Delivered Orders	COUNT where Status=Delivered	2,34,461	Fulfilment performance
Revenue by Segment	SUM(Total_Amount) by Segment	Varies	Segment profitability
Total Products Sold	SUM(Total_Purchases)	1,616,727	Volume of product movement across all categories
Total Brands	COUNTD(Product_Brand)	18	Brand portfolio breadth and diversity

7. Exploratory Data Analysis

Revenue Trends

Monthly revenue analysis reveals a strong peak in April (Rs. 60Cr) followed by a consistent decline through the second half of the year, reaching a low in December. This pattern suggests strong Q1 performance driven by seasonal buying behaviour, with Q4 representing a significant opportunity for targeted promotional campaigns. The weekday vs weekend analysis shows weekdays generating Rs. 296Cr (71.7%) compared to weekends at Rs. 117Cr (28.3%), indicating the importance of B2B or professional buyer segments.

Customer Segmentation

The customer base is dominated by Regular customers (48.47% by revenue, Rs. 200Cr) followed by New customers (30.20%, Rs. 125Cr) and Premium customers (21.25%, Rs. 88Cr). This distribution reveals an opportunity: Premium customers

generate high individual value but represent only 21% of revenue, suggesting an upgrade programme could significantly increase total revenue. The age distribution shows 26-35 year olds as the largest segment (28.7%), with Adults being the dominant group overall.

Product Performance

Product category analysis shows Grocery and Clothing leading by revenue, followed by Books and Home Decor. Electronics is the top product category overall with 1,616,727 total units sold across 18 unique brands. The product type treemap reveals Water, Smartphone and Fiction as the highest-revenue product types. Top brands by revenue include Samsung, Pepsi, Sony and Zara, while BlueStar and Mitsubishi lead in average customer ratings (above 4.0), indicating strong product quality perception in those brands.

Operational Insights

Order status distribution shows Delivered as the largest category (130K orders, 43%), followed by Shipped (65K, 22%), Processing (57K, 19%), and Pending (49K, 16%). The 16% pending rate represents a significant operational risk. Feedback distribution reveals 66% positive, 25% neutral, and 9% negative feedback, indicating generally high customer satisfaction.

8. Statistical Analysis

Hypothesis Testing

T-Test (Premium vs Regular Customer Spending): The analysis tested whether Premium customers spend significantly more than Regular customers. Result: statistically significant difference ($p < 0.05$), confirming Premium customers have a meaningfully higher average order value. This validates the business case for a Premium customer acquisition programme.

Chi-Square Test (Gender vs Product Category): The test examined whether gender is independent of product category preference. Result: statistically significant relationship ($p < 0.05$), indicating gender-based product preferences exist. This supports gender-targeted marketing campaigns.

ANOVA Test (Spending across Product Categories): One-way ANOVA tested whether mean spending differs across product categories. Result: statistically significant ($p < 0.05$), confirming different categories attract different spending levels.

RFM Segmentation & KMeans Clustering

RFM (Recency, Frequency, Monetary) analysis was performed on customer data with StandardScaler applied to normalise the three dimensions before KMeans clustering with k=3. The resulting segments were classified as High Value (customers

with recent, frequent, high-value transactions), Mid Value (moderate engagement), and Low Value (infrequent, lower-spend customers). High Value customers represent the primary retention target.

Revenue Forecasting

Linear regression was applied to monthly revenue data to identify the trend direction. The model captures the overall revenue trajectory, though it does not account for seasonality. The forecast serves as a conservative baseline for revenue planning. A more sophisticated ARIMA or seasonal decomposition model is recommended for future analysis.

9. Tableau Dashboard Design

- All four dashboards follow a consistent design template:
  - Dark header banner
  - KPI cards in the top row
  - Interactive charts below
- **Color Theme**:
  - Primary Accent: Orange (**#f97316**)
  - Secondary: Grey (**#78716c**)
  - Background: Warm Cream (**#f5f5f5**)

Dashboard 1: Sales & Revenue Overview

- **Objective**:
  - Enable executive-level monitoring of overall business performance
- **KPI Cards**:
  - Total Revenue
  - Total Orders
  - Average Order Value (AOV)
  - Average Rating
- **Charts Included**:
  - Monthly Revenue Trend (Line Chart)
  - Revenue by Customer Segment (Bar Chart)
  - Revenue by Country (Treemap)
  - Shipping Method (Bar Chart)
  - Payment Method (Bar Chart)
  - Order Status (Pie Chart)
  - Weekend vs Weekday Sales (Bar Chart)
- **Filters**:
  - Country
  - Order Status
  - Customer Segment
  - Month

Dashboard 2: Customer Insights Overview

- **Objective**:
  - Support marketing and CRM decisions using customer intelligence
- **KPI Cards**:
  - Total Unique Customers
  - Average Customer Age
  - Top Customer Segment
  - Retention Rate
- **Charts Included**:
  - Age Distribution (Donut Chart)
  - Customers by Gender (Donut Chart)
  - Customers by Segment (Treemap)
  - Customer Acquisition Trend (Line Chart)
  - AOV by Age Group (Bar Chart)
  - Income Distribution (Pie Chart)
  - Customer Value Segment (Bubble Chart)
- **Filters**:
  - Customer Segment
  - Month
  - Age Group

Dashboard 3: Product Performance Analysis

- **KPI Cards**:
  - Top Product Category: Electronics
  - Total Products Sold: 1,616,727
  - Average Rating: 3.163
  - Total Brands: 18
- **Charts Included**:
  - Revenue by Product Category (Bar Chart – Grocery & Clothing leading)
  - Revenue by Purchase Category (Pie Chart – Medium vs Low)
  - Top 10 Brands by Revenue (Bubble Chart)
    - Samsung, Pepsi, Sony, Zara, Nike, Adidas, Coca-Cola, HarperCollins, Penguin Books, Bed Bath & Beyond
  - Revenue by Product Type (Treemap – Water, Smartphone, Fiction highest)
  - Top 5 Brands by Ratings (Bar Chart – BlueStar & Mitsubishi leading)
- **Filters**:
  - Product Category
  - Product Brand
  - Product Type

Dashboard 4: Operational Insights Overview

- **Objective**:
  - Monitor order fulfillment and operational efficiency
- **KPI Cards**:
  - Order Success Rate
  - Total Delivered Orders
  - Most Used Shipping Method
  - Average Rating

- **Charts Included**:
  - Order Status Distribution
  - Revenue by Shipping Method (Donut Chart)
  - Feedback Distribution
  - Order Success by Customer Segment
  - Revenue by Customer Value
  - Orders by Weekday
- **Filters**:
  - Order Status
  - Shipping Method
  - Customer Segment
  - Month

10. Insights Summary

#	Insight	Business Implication
1	Regular customers contribute 48% of revenue	Retention is more valuable than acquisition — invest in loyalty programmes
2	April is peak revenue month at Rs. 57 Cr	Inventory and staffing must be scaled up in Q1 to maximise peak season revenue
3	December revenue is the lowest of the year	Targeted promotions and discounts in Dec could recover Rs. 8-10Cr
4	Same-Day shipping generates Rs. 142Cr — highest of all methods	Expand same-day delivery capacity to drive further revenue growth
5	94.7% order success rate	The 5.3% failure rate represents ~Rs. 22Cr in recoverable revenue
6	16% of orders are pending	Process bottleneck requiring operational review to prevent customer churn
7	66% positive feedback rate	Strong baseline satisfaction; targeting neutral customers (25%) could boost NPS significantly
8	Premium customers spend significantly more (t-test confirmed)	Premium upgrade programmes have strong statistical basis for investment
9	Gender significantly influences product preferences (chi-square)	Gender-targeted marketing will outperform generic campaigns
10	26-35 age group is largest customer segment	Marketing messaging should be optimised for early-career digital-native consumers
11	Weekdays generate 71.7% of revenue	B2B promotional strategies should focus on weekday engagement
12	Credit Card is highest-revenue payment method	Partnerships with credit card providers could unlock loyalty rewards integration

11. Recommendations

Recommendation 1: Launch Regular Customer Loyalty Programme

Insight: Regular customers drive 48% of revenue. Recommendation: Introduce a points-based loyalty programme with exclusive discounts for Regular segment customers. Expected Impact: 10-15% increase in repeat purchase rate, estimated Rs. 20-30Cr additional annual revenue. Feasibility: Moderate — requires CRM integration and marketing budget allocation.

Recommendation 2: December Promotional Campaign

Insight: December is the lowest revenue month. Recommendation: Deploy targeted discount campaigns and bundle offers in November-December to stimulate demand. Expected Impact: 15-20% revenue uplift in December, recovering Rs. 8-10Cr. Feasibility: Low complexity — existing marketing channels can be used immediately.

Recommendation 3: Expand Same-Day Delivery

Insight: Same-Day shipping generates the highest revenue (Rs. 142Cr). Recommendation: Negotiate expanded same-day delivery partnerships with logistics providers to cover more regions. Expected Impact: 10-12% increase in Same-Day adoption, estimated Rs. 14-17Cr incremental revenue. Feasibility: Moderate — requires logistics partnership and warehousing investment.

Recommendation 4: Resolve Pending Orders Bottleneck

Insight: 16% of orders remain pending. Recommendation: Implement automated order processing triggers and SLA monitoring to reduce pending orders. Expected Impact: Reduction of pending rate from 16% to 8%, recovering Rs. 10-15Cr in at-risk revenue. Feasibility: High — can be implemented with existing order management systems.

Recommendation 5: Premium Customer Upgrade Programme

Insight: Premium customers spend significantly more (statistically confirmed). Recommendation: Create a Regular-to-Premium upgrade pathway with personalised offers. Expected Impact: 5% conversion of Regular to Premium customers adds Rs. 10Cr annually. Feasibility: Moderate — requires targeted communication and offer design.

12. Impact Estimation

Recommendation	Estimated Impact	Timeframe	Confidence
Loyalty Programme	+Rs. 20-30Cr annual revenue	6-12 months	High
December Campaign	+Rs. 8-10Cr seasonal revenue	1-2 months	High

Recommendation	Estimated Impact	Timeframe	Confidence
Same-Day Expansion	+Rs. 14-17Cr annual revenue	6-12 months	Medium
Pending Orders Fix	Recover Rs. 10-15Cr at-risk	1-3 months	High
Premium Upgrade	+Rs. 10Cr annual revenue	3-6 months	Medium
Total Potential	+Rs. 62-82Cr annually	12 months	Medium-High

13. Limitations

- The dataset covers one fiscal year only; multi-year trend analysis and year-over-year comparisons are not possible.
- Geographic granularity is limited to country level; regional sub-national patterns cannot be identified.
- PII removal (Name, Email, Phone) prevents individual customer-level personalisation analysis.
- The revenue forecasting model uses linear regression which does not capture seasonality; actual future revenue may deviate significantly.
- Null values filled with median/mode introduce slight statistical bias in affected columns.
- The dataset source and collection methodology are not fully documented, limiting reproducibility assessment.

14. Future Scope

- Build a machine learning churn prediction model using customer RFM scores and order history to identify at-risk customers before they lapse.
- Integrate real-time data ingestion using an API pipeline to convert the Tableau dashboard from static to live.
- Incorporate external data sources such as competitor pricing, market trends, and economic indicators to contextualise revenue performance.
- Develop a demand forecasting model using ARIMA or Prophet to predict monthly revenue and support inventory planning.
- Expand geographic analysis by integrating city and state-level data for regional marketing optimization.

15. Conclusion

In an increasingly competitive retail landscape, extracting actionable intelligence from raw data is critical for strategic growth and operational efficiency. This project addressed the challenge of opaque customer behaviors and operational bottlenecks by transforming 302,010 raw transactional records into a robust, end-to-end analytics pipeline utilizing a structured five-stage Python ETL process and four interactive Tableau dashboards.

Through rigorous exploratory and statistical analysis, we uncovered that Regular customers form the financial backbone of the business by contributing 48% of total revenue, while also identifying significant untapped potential in mitigating Q4 seasonal revenue dips and capitalizing on the high demand for Same-Day shipping.

To convert these insights into measurable business value, we strongly recommend the immediate implementation of a targeted Regular Customer Loyalty Programme, a strategic December promotional campaign, and the geographical expansion of Same-Day delivery infrastructure. Ultimately, this comprehensive analytics system empowers executive stakeholders with the clear, data-backed visibility needed to optimize operations, enhance customer retention, and confidently drive an estimated Rs. 62-82Cr in additional annual revenue.

Appendix

A. Data Dictionary (Key Engineered Columns)

Column	Values	Description
age_group	18-25, 26-35, 36-45, 46-60, 60+	Customer age bracket
purchase_category	Low / Medium / High / Very High	Based on Total_Purchases count
revenue_category	Low / Medium / High / Premium	Based on Total_Amount value
is_weekend	True / False	True if Saturday or Sunday
order_success	0 or 1	1 = Delivered, 0 = Otherwise
rating_category	Poor / Average / Good / Excellent	Based on Ratings score
customer_value_segment	Low / Medium / High / Very High Value	Based on Total_Amount
avg_order_value	Float	Total_Amount / Total_Purchases

B. GitHub Repository Structure

SectionB_Group3_RetailAnalytics/

- data/raw/ — Original dataset (never edited)
- data/processed/ — cleaned_dataset.csv and tableau_ready_dataset.csv
- notebooks/ — 01_extraction to 05_final_load_prep
- scripts/etl_pipeline.py — Encapsulated ETL pipeline
- tableau/screenshots/ — Dashboard screenshots
- tableau/dashboard_links.md — Tableau Public URL
- reports/figures/ — KPI chart exports
- docs/data_dictionary.md — Full column definitions

18. Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Bulbul Agarwalla	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Shourya Pratap	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nishant Sharma			Yes	Yes			Yes
Udata Varun						Yes	Yes
Anant Lalwani		Yes					Yes
Parul	Yes						Yes

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: Bulbul Agarwalla Date: April 29, 2026